Realistic deepfakes are detected better from a single glance than deliberate inspection

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Competing interest

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Main text (word count: 1794)

Figure 1

Abstract

The rapidly increasing realism of AI-generated faces threatens trust in digital media, as observers often fail to reliably distinguish AI-generated faces from real faces under prolonged viewing. Across three experiments, participants (N = 300) classified faces varying in realism, exposure duration, and feature availability as real or AI-generated. Observers detected AI-generated faces above chance based on 83 ms when external features, such as hair and background, were present, whereas unlimited viewing did not improve detection of realistic AI-generated faces. These findings suggest observers can extract gist-based authenticity-related signals from a single glimpse, but these may be overridden during prolonged evaluative face processing.

Introduction

Advances in AI-generated content have rendered synthetic media near-impossible for humans to detect under normal viewing conditions (1). The alarming misuse of deepfakes has intensified efforts to understand how well humans can identify deepfakes, revealing that people often fail to recognise synthetic faces and are overconfident in their errors (2, 3). Interventions and training provide limited benefits (4, 5). Thus far, deepfake detection has been examined in the context of unrestricted viewing (2). Here we considered whether detailed inspection of deepfakes might enable higher-order cognitive evaluations that override initially-accurate perceptual cues signaling deepfake status. We reasoned that limiting viewing duration would constrain reflective processing, instead promoting rapid, gist-based judgments based on fast, coarse impressions extracted from early visual processing (6). Face processing is well-known to operate rapidly, with key face information extracted within 100 ms of viewing (7, 8). Moreover, implicit neural discrimination of real and AI faces occurs even when explicit behavioural classification is at chance (9). This dissociation suggests that the visual system registers distinctions between real/fake faces rapidly, yet whether such early neural distinctions can be accessed to support explicit deepfake detection remains unknown. Furthermore, it is unclear how brief exposure influences the use of external face aspects during deepfake detection. Here, we manipulated exposure duration and external-feature availability to examine how gist-based processing supports human performance in detecting AI-generated faces. The same deepfakes that could not be detected above chance during unlimited viewing were discernible from real faces when presented very briefly.

Results

Participants in three online studies (each N=100) classified face singletons as real or fake. Stimuli comprised real, unrealistic AI-generated, and realistic AI-generated faces, presented at various exposure durations, both with and without external features (Fig 1A).

Observers detect realistic AI-generated faces based on a single glance (Fig 1D). When external features were available (e.g., hair/background), observers reliably discriminated realistic AI-generated faces from real faces at an exposure duration of 83 ms, $M_{d'}$ = 0.27, *SD* = 0.73, *t*(99) = 3.62, p_{bonf} = 0.002, *d* = 0.36, BF_{10}= 44.60. Unrealistic AI-generated faces with external features were detectable at 66 ms, $M_{d'}$= 0.18, *SD* = 0.63, *t*(99) = 2.81, p_{bonf} = 0.03, *d* = 0.28, BF_{10} = 4.49.

External features support deepfake detection at brief durations. Across brief exposures, discrimination was higher when external features were available than when only internal face features were preserved, *F*(1, 198) = 20.83, *p* < 0.001, ηp^2 = 0.095, BF_{10} = 1.00. Critically, when external features were removed, observers could no longer discriminate realistic AI-generated faces from real faces above chance across all exposure durations. Thus, rapid detection of realistic AI-generated faces appears to rely on information carried by aspects external to the face itself.

Unrestricted viewing does not improve detection of realistic AI-generated faces. When viewing time was unrestricted, both stimulus-realism and external-feature presence modulated detection performance, F(1, 99) = 15.93, *p* < 0.001, ηp^2 = 0.139, BF10 = 0.007. Observers reliably detected unrealistic AI-generated faces regardless of whether external features were present, $M_{d'}$ = 0.42, SD = 0.36, t(99) = 11.49, *p* < 0.05, d = 1.15, BF_{10} = 7.53 x 10^{16}, or removed, $M_{d'}$ = 0.36, SD = 0.35, t(99) = 10.08, *p* < 0.05, d = 1.00, BF_{10} = 8.14 x 10^{13} (Fig. 1D-E, orange dashed lines). By contrast, realistic AI-generated faces were not reliably distinguished from real faces when external features were available, $M_{d'}$ = -0.04, SD = 0.64, t(99) = -0.56, *p* = 0.579, d = -0.06, BF_{10} = 0.129 (Fig. 1D, blue solid line), with Bayesian evidence favouring chance-level performance. Notably, discrimination of realistic AI-generated faces dropped below chance when external features were removed, $M_{d'}$ = -0.29, SD = 0.62, t(99) = -4.78, *p* < 0.05, d = -0.48, BF_{10} = 2414.90, indicating that observers systematically misclassified realistic AI-generated faces as real when only internal face features were available. Thus, unrestricted viewing did not improve detection of realistic AI-generated faces, suggesting that the sensitivity observed under brief exposure was lost during close inspection.

Confidence does not track detection performance. At brief exposures, observers' confidence in their detection performance varied with stimulus-realism, F(1, 198) = 22.51, *p* < 0.001, ηp^2 = 0.102, and exposure duration, F(4, 792) = 203.19, *p* < 0.001, ηp^2 = .506, but not feature availability, F(1, 198) = 1.31, *p* = 0.253, ηp^2 = 0.007. Feature availability did not interact with stimulus-realism or exposure-duration (all *ps* > 0.05). Under unlimited viewing, participants' confidence varied with external-feature availability and stimulus-realism, Fs ≥ 5.51, *ps* ≤ 0.005, but was descriptively highest for realistic faces with internal features (*M* = 2.904, SD = 0.605), the condition showing the strongest detection misclassification ($M_{d'}$ = -0.29, SD = 0.62). Thus, observers were most confident when they were least accurate.

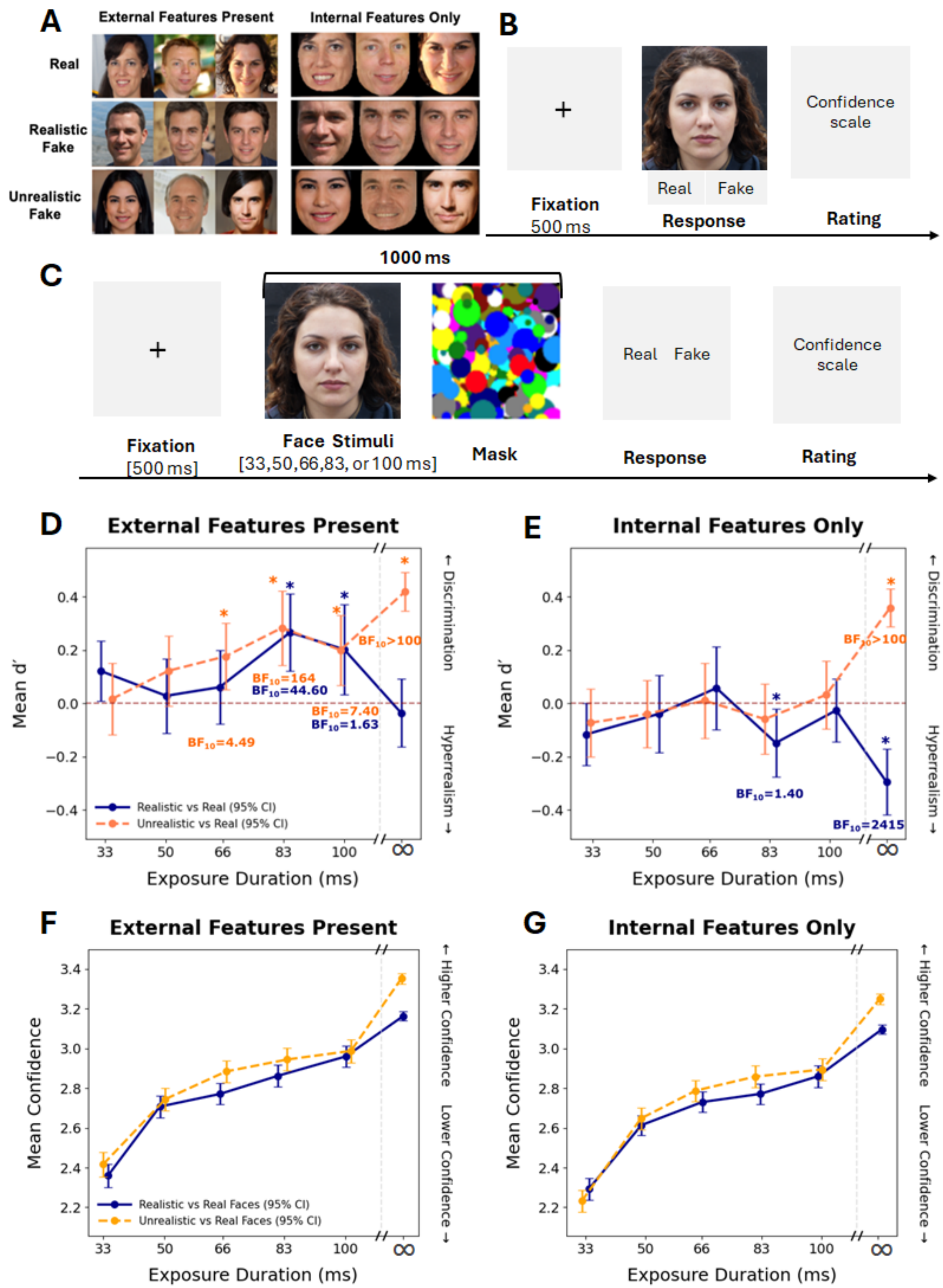

Fig 1. (A) Real, realistic AI-generated, and unrealistic AI-generated faces shown with and without external features. (B) Study 1: Observers classified faces as real/fake based on unlimited exposure, followed by confidence rating. (C) Studies 2a–2b followed the same procedure, save that faces appeared for 33, 50, 66, 83, or 100 ms and were immediately masked. (D–E) Mean discrimination sensitivity (d′) across exposure duration when external features were preserved (D) or removed (E). (F–G) As above, for mean confidence ratings. Error bars show 95% confidence intervals. Dashed horizontal lines indicate chance performance (d′ = 0). The final point in D-G is the unlimited viewing condition.

Discussion

A brief glimpse can support detection of realistic AI-generated faces

Our data provide the first evidence that observers can reliably detect AI-generated faces based on a single brief glimpse. Where extant work suggests humans struggle to detect deepfakes when allowed to view faces freely (2, 3), observers here discriminated realistic AI-generated faces from real faces based on just 83 ms of exposure. This finding implies that where people are generally poor at identifying deepfakes, detection capabilities may be best supported by early, gist-based (6) processing over deliberate or strategy-based inspection.

External features carry the most diagnostic information

Above-chance detection following brief exposures was not maintained when observers did not have access to external features. The at-chance performance in the internal-features-only condition is broadly consistent with evidence that external face information contributes substantially to face processing and person judgments (10). A stimulus-driven account of this finding posits that generative-models may produce subtle inconsistencies in peripheral regions like hair, making these cues informative for detection (11). Alternatively, external features may support the rapid global face representations observers rely on during gist-based processing (7). Future work could dissociate these possibilities by selectively manipulating the fidelity or availability of specific external features.

Longer viewing worsens detection of realistic AI-generated faces

Contrary to official advice to “look closer” (12), we found allowing unlimited exposure reduced detection of realistic AI-generated faces to at or below chance-level. Prolonged inspection may engage higher-level face processing mechanisms that prioritise holistic identity and familiarity, overriding sensitivity to subtle authenticity-related cues in AI-generated faces. This interpretation is consistent with prior work reporting chance-level performance under longer or unrestricted viewing conditions (2, 3). Nevertheless, detection may depend on a brief processing window in which the visual system can extract useful authenticity-related information (9), consistent with evidence that rapid face categorisation occurs within early visual processing (13).

Early detection reflects gist extraction and reduces hyperrealism

Face perception unfolds rapidly, with meaningful information available within the first 100 ms of viewing (7, 14). Under brief exposures, observers may rely on global gist-like signals before slower more effortful processes emerge (15). With unlimited viewing, however, responses shifted towards classifying realistic AI-generated faces as real, consistent with a ‘hyperrealism’ bias in which synthetic faces appear highly prototypical (16, 17). The data here suggest that where unlimited viewing may amplify this bias, constrained viewing time may help preserve access to early authenticity-related signals that are rapidly overridden during deliberate inspection.

Confidence in deepfake detection is dissociated from performance

Notable here is that subjective confidence in deepfake detection did not track objective accuracy: Participants remained relatively confident despite poor or systematically incorrect detection of realistic AI-generated faces, suggesting metacognitive miscalibration consistent with a Dunning–Kruger-like pattern (18). This aligns with prior work showing that realistic synthetic faces can encourage a “seeing-is-believing” heuristic, whereby observers may accept visually realistic faces at face value unless obvious cues reveal them as fake (3).

Conclusion

Human detection of AI-generated faces is possible, but highly conditional. Observers can extract authenticity-related information from a single gist-based glimpse, particularly when external facial information is available. This suggests that real/fake differences may emerge early in visual processing and can be lost during prolonged evaluation. These findings challenge current deepfake detection advice that assumes longer inspection improves accuracy.

Materials and Methods

Three online experiments were conducted in JsPsych/Pavlovia with Prolific participants from Australia and New Zealand (N = 100 per experiment). Stimuli were 300 real FFHQ faces, 300 realistic StyleGAN2 faces, and 300 unrealistic StyleGAN faces, screened for quality, neutrality, frontal pose, and artifacts, then downsampled to 128x128 pixels. Faces were shown with external-features-present or internal-features-only using OpenFace masking. Study 1 used unlimited viewing across 300 trials; Studies 2a/2b used 33-100 ms masked presentations across 150 trials. Detection was analysed using d′, t tests, and repeated-measures ANOVAs. Confidence was analysed with corresponding ANOVAs.

Ethics

This study was approved by Western Sydney University Human Research Ethics Committee (Ethics code: H14498). All participants provided informed consent prior to participation, and all procedures were conducted in accordance with relevant institutional guidelines and regulations.

Data availability

Anonymised behavioural data, analysis code, and experimental materials will be made publicly available on OSF.

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Supplementary Materials and Methods

Participants

We recruited 300 participants through Prolific across three online experiments, with 100 participants per experiment. Recruitment was restricted to participants located in Australia and New Zealand. Eligibility criteria required participants to be fluent in English, have normal or corrected-to-normal vision, and report no history of epilepsy. All participants provided informed consent prior to participation. In the unlimited exposure experiment (study 1), which included both external-feature-present and internal-features-only stimuli, participants ranged in age from 19 to 74 years (Mdn = 34, *SD* = 12.78, 60 females, 40 males). In the brief-exposure experiment with external features present (study 2a), participants ranged in age from 18 to 79 years (Mdn = 26.5, *SD* = 12.45; 52 females, 48 males). In the brief-exposure experiment with internal features only (study 2b), participants ranged in age from 20 to 79 years (Mdn = 42, *SD* = 13.49; 49 females, 51 males).

Apparatus

The three experiments were conducted online using the JsPsych v7.3 library and hosted on Pavlovia.org. To ensure adequate control over the stimulus exposure duration, the task had to be completed on a laptop and desktop computer with a refresh rate of at least 60 Hz. The experiment script blocked participants who did not meet these requirements.

Stimuli

The visual stimuli used in this study were obtained from three established databases: the Flickr-Faces-HQ (FFHQ), StyleGAN and StyleGAN2. These datasets were selected due to their high image quality, photorealism, and frequent use in previous research on face perception and generative model evaluation. The final stimulus set comprised 300 real faces from FFHQ, 300 unrealistic AI-generated faces generated using StyleGAN, and 300 realistic AI-generated faces generated using StyleGAN2. StyleGAN2 faces were labelled “realistic” because StyleGAN2 improves image quality and reduces artifacts relative to StyleGAN, producing faces that are more visually convincing and harder to detect as fake. The StyleGAN face set was generated with truncation parameter $\psi = 1.0$, meaning that no truncation was applied, whereas the StyleGAN2 face set was generated with $\psi = 0.5$. The truncation parameter controls how strongly generated faces are constrained toward more typical or average-looking faces, with lower values generally producing more regular but less variable images. All images were screened to remove duplicates, obvious visible distortions, and other artifacts. To minimise superficial differences between real and fake images, we excluded faces with non-frontal gaze, exaggerated emotional expressions, accessories, or additional people in the background. Only neutral, centrally aligned faces were retained. The final set was balanced for apparent demographic characteristics, including gender, age, and ethnicity, to ensure comparability across conditions.

Stimuli were restricted to Caucasian faces. This decision was motivated by evidence that current generative face models are disproportionately trained on datasets containing predominantly Caucasian identities, which can produce higher realism and greater detection difficulty for these faces. Restricting the stimulus set therefore helped ensure that differences in detection performance reflected the experimental manipulations rather than variability in image quality across racial groups. In addition, categorised non-Caucasian image sets were not available within the stimulus database, and manually assigning racial categories would have introduced potential classification biases. To reduce storage requirements and support efficient online delivery, all images were down sampled from 1024x1024 to 128x128 pixels. We created two stimulus versions: external-feature-present stimuli and internal-features-only stimuli. External-feature-present stimuli retained the full image context, including the face outline, hair, ears, and background. Internal-features-only stimuli were generated using OpenFace, to extract face landmarks and mask non-face regions. Stimuli were selected through automated preprocessing followed by manual curation. First, OpenFace was used to extract face landmarks and standardise alignment. All stimuli were then manually reviewed to ensure that neutral expression, alignment, and image quality criteria were met. We retained correspondence between each internal-features-only image and its external-feature-present counterpart, ensuring that both versions of each stimulus were free from obvious confounds.

Design

Study 1 (unlimited duration) employed a 2x2 factorial design with Presentation Type (external-feature-present face image vs. internal-feature face image) and Stimulus Realism (realistic AI-generated faces (fake), unrealistic AI-generated faces (fake)) as within-subject factors. Stimuli in study 1 were drawn from a pool of 900 face images, comprising of 300 real, 300 realistic AI-generated, and 300 unrealistic AI-generated faces. Each identity was presented in external-feature-present and internal-features-only versions. For each participant, 50 face images of each stimulus type (real face, realistic AI-generated face and unrealistic AI-generated face) were randomly sampled from this pool, yielding a 150-image stimulus set. Both versions of each selected identity (external features present and internal features only) were then included in the experiment, resulting in 300 trials. Stimuli were presented in randomised order across experimental blocks.

Study 2a and 2b, are distinct only in the way stimuli were presented, either with external-features present (study 2a) or internal-features only (study 2b), both also using randomized exposure paradigms (33, 50, 66, 83, 100 ms). By systematically varying exposure duration, we increased reliance on rapid perceptual processing thereby revealing the temporal limits and potential capabilities of the visual system in distinguishing real from fake faces. Here, stimulus type (real, realistic, unrealistic) was randomly assigned to exposure durations, resulting in approximately equal numbers of each stimulus type at every duration.

Procedure

Participants first provided informed consent then completed some demographic questions. Participants then completed a screen-size calibration procedure so that stimuli could be presented at a consistent physical size across devices. Each experiment began with an instruction screen followed by 15 practice trials that mirrored the structure of the main task and included the relevant stimulus conditions. During practice, participants received feedback on whether their real/fake response was correct. After completing the practice trials, participants viewed a second instruction screen before beginning the main task. On each trial, participants viewed a face and judged whether it was “real” or “fake.” The position of the “real” and “fake” response buttons switched between blocks. After this response, participants rated their confidence (“How confident are you in your response?”) using a vertically presented Likert scale with the following options, from top to bottom: “Very Confident,” “Confident,” “Slightly Confident,” “Not Confident,” and “Did not see a face.” The main task followed the same response procedure as the practice trials, except that participants did not receive feedback. At the end of each experiment, participants completed a short text-response survey in which they described the strategies or cues that influenced their real/fake judgments, prompted by the question: “What factors influenced your decision about whether each face was real or fake? (e.g., face features, lighting, intuition, etc.)”

In the unlimited-exposure experiment (study 1), faces remained on screen until participants responded. Participants (N = 100) completed 300 main trials across 30 blocks, with a break after every 10 trials. This experiment included both external-feature-present and internal-features-only stimuli.

Study 2a and Study 2b followed the same general response procedure as study 1 but used brief exposure durations and backward masking (see figure 1b and 1c). In Study 2a, participants (N = 100) viewed faces with external features available, whereas in Study 2b, participants (N = 100) viewed only internal face features against a black background (see figure 1a). In both studies, faces were presented for 33, 50, 66, 83, or 100 ms. Immediately after each face, a visual mask consisting of coloured circles was presented at the same size as the face stimulus. Mask duration varied according to face duration so that the combined face-and-mask display lasted 1000 ms. After the face and mask were presented, participants judged whether the face was real or fake and then rated their confidence. Each participant completed 150 main trials across five blocks, with a break after every 30 trials.

Analysis

Detection performance was indexed using *d′* (sensitivity), where higher values indicate better discrimination between signal and noise. d' was quantified using signal detection theory (Green & Swets, 1966). Study 1 was analysed using a 2 × 2 repeated measures analysis of variance (ANOVA) with external/internal features and stimulus realism (realistic *vs*. real and unrealistic *vs.* real) as factors. to analyse confidence ratings, a 2 x 2 ANOVA was conducted to examine the effects of external/internal features and stimulus type, as within-subject factors, allowing us to test for main effects and interactions. For study 2a (external features present) and 2b (internal features only), one-sample *t*-tests were used to compare mean *d′* values against chance performance (*d′* = 0) at each exposure duration. Both uncorrected and Bonferroni-corrected *p* values were reported, alongside Cohen's *d* effect sizes and Bayes factors (BF_{10}). To assess overall effects across brief exposures, *d′* was also analysed using a mixed ANOVA with study (Study 2a: external features present; Study 2b: internal features only) as a between-subjects factor, and stimulus realism and exposure duration (33, 50, 66, 83, and 100 ms) as within-subjects factors. Confidence ratings were analysed using the same mixed-ANOVA structure.